

Felipe Operti

AI Architect | Data Scientist | ML Engineer

Turin, Italy

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in felipeoperti

AI Architect and Data Science Specialist with deep expertise in NLP, Generative AI, and scalable ML systems. I design, develop, and deploy end-to-end AI solutions across multiple cloud environments, lead data science teams, and turn complex business challenges into production-ready models. With experience spanning insurance, banking, technology, and research, I combine strong engineering skills with strategic vision to build reliable and innovative AI products. Fluent in English, Italian, and Portuguese.

EMPLOYMENT

AI Architect

Reale Mutua

Turin, Italy

Oct 2025 - Present

As an AI Architect, I design and develop advanced AI solutions to support business goals. My role focuses on defining end-to-end AI/ML solutions, as well as coordinating and supporting the work of the data science team.

Technologies: Artificial Intelligence (AI), Natural Language Processing (NLP), Generative AI, Computer Vision, Machine Learning, Solution Architecture, Management, Databricks, and Azure.

Data Science Specialist

Reale Mutua

Turin, Italy

Jul 2022 - Oct 2025

As a Data Science Specialist at Reale Mutua, my primary focus was on working with unstructured data using Natural Language Processing (NLP) and deep learning models. I worked on Azure, leveraging its capabilities to design, develop, and deploy advanced AI solutions.

Technologies: Generative AI, Azure, AzureML, Databricks, LangGraph, LangChain, Python, Docker, PyTorch, HuggingFace, Spacy, etc.

Data Scientist Specialist | Machine Learning Engineer

Self-employed

Turin, Italy

Jan 2021 - Present

FraleAnalitica was born from a personal project with a friend, aiming to apply machine learning and deep learning techniques in agriculture, particularly beekeeping. At FraleAnalitica, I work as a technology consultant, freelance data scientist, and machine learning engineer. My primary responsibilities include developing, deploying, and maintaining AI services for agricultural and beekeeping applications.

For more information:

- <https://opertifelipe.github.io/projects/honai/>

- <https://www.fraleanalitica.it/>

Technologies: Azure, MLOps, Machine Learning, Deep Learning, FastAPI, Docker, Python, etc.

Data Scientist | Machine Learning Engineer

Inda - Interviewweb S.r.l. - Zucchetti Group

Turin, Italy

Mar 2021 - Jul 2022

Data Scientist at INDA (Interviewweb S.r.l., a Zucchetti company), working mainly with Natural Language Processing (NLP) and deep learning. Responsible for the development, deployment, and maintenance of machine learning solutions.

Technologies: AWS EKS, AWS ECS, AWS Lambda, AWS SageMaker, Python, Docker, Kubernetes, FastAPI, Elasticsearch, MongoDB, etc.

Data Scientist Specialist

Samsung SDS

São Paulo, São Paulo, Brazil

Nov 2020 - Mar 2021

Data Science Specialist in the CRM area at Samsung SDS. Responsible for the development and deployment of propensity models, churn models, sentiment analysis, and RFV models.

Technologies: Data Lake, HDFS, Brightics, Adobe Campaign, Python, Scala, Docker, Spark, and PySpark.

Data Scientist Consultant

Santander Bank

São Paulo, São Paulo, Brazil

Aug 2019 - Jul 2020

Data Scientist in Santander DataLab. DataLab is an innovative unit focused on developing large, cross-functional end-to-end projects. DataLab works as an internal consultancy within the bank. In the lab, the interaction between product owners, data engineers, data scientists, and machine learning engineers is strongly encouraged. My responsibilities were twofold: on one hand, I developed machine learning models as a data science consultant; on the other hand, I was responsible for validating the end-to-end pipeline (from ETL and model development to deployment). *Technologies:* Data Lake, Hadoop, Cloudera, Python, Scala, SQL, Hive, Impala, Spark, PySpark, and Oracle.

Senior Data Scientist

São Paulo, São Paulo, Brazil

Santander Bank

Dec 2018 - Aug 2019

Senior Data Scientist at Santander Bank in Customer Relationship Management (CRM), developing machine learning models for multiple use cases (e.g., propensity and churn). These models typically used supervised algorithms such as Random Forest, Gradient Boosting, Logistic Regression, and Neural Networks, as well as unsupervised methods such as hierarchical clustering, k-means, and DBSCAN.

Technologies: Data Lake, Hadoop, Cloudera, Python, R, Scala, SQL, Java, ShellScript, Hive, Impala, Spark, PySpark, and Oracle.

EDUCATION

PhD in Physics, Complex Systems

Fortaleza, Brazil

Federal University of Ceará

Feb 2015 - Dec 2018

Thesis: *Computational analysis for socio-economic sciences.*

Visiting PhD in Physics, Complex Systems

New York City, United States

City College of New York

Apr 2018 - Jun 2018

Visiting PhD in Physics, Complex Systems

Rome, Italy

University La Sapienza

Apr 2017 - Sep 2017

MSc in Physics, Complex Systems

Fortaleza, Brazil

Federal University of Ceará

Jul 2014 - Jan 2015

Thesis: *Interpolation strategy based on Dynamic Time Warping.*

BSc in Physics

Turin, Italy

University of Turin

Sep 2009 - Oct 2012

Thesis: *Monte Carlo Simulation of the radiation distribution emitted by a CT scan in the field of the radioprotection.*

SKILLS

Machine Learning and Deep Learning: LangGraph, LangChain, PyTorch, Unsloth, HuggingFace, OpenAI, Azure Cognitive Services, MCP, A2A, Spacy, etc.

MLOps: FastAPI, Kubernetes, Docker, Git, Typer, Pydantic, Poetry, uv, GitHub Actions, Azure Pipelines, etc.

Cloud: Azure, AzureML, Azure DevOps, Azure Container App, Databricks, AWS, AWS ECS, AWS EKS, AWS ElasticBeanstalk, AWS SageMaker, etc.

Data Analysis: Python (Jupyter Notebook, Pandas, NumPy, StatsModels, etc), Scala, SQL, Spark, PySpark, Hadoop, Hive, Impala, Excel, etc.

Data Visualization: Matplotlib, Seaborn, Plotly, Streamlit, Grafana, Gephi, QGIS, etc.

Databases: MySQL, PostgreSQL, HDFS, Elasticsearch, and MongoDB.

Other computer skills: Java, C++, (learning) Rust, and Bash.

Operating systems: Linux (Arch Linux, Debian, and Ubuntu), macOS, and Windows.

Languages: Italian (native), Portuguese (fluent), and English (fluent).

PUBLICATIONS

- Felipe G. Operti et al. Dynamics of racial segregation and gentrification in New York City. Frontiers in Physics. 2022.
- Felipe G. Operti et al. Dynamics in the Fitness-Income plane: Brazilian states vs World countries. PLOS ONE. 2018.
- Felipe G. Operti et al. The light pollution as a surrogate for urban population of the US cities. Physica A. 2018.